

Towards Cassels-Swinnerton-Dyer conjecture for cubic threefolds

Abstract

We study rational points on smooth cubic threefolds over fields of characteristic zero. Taking degree-five descent for smooth cubic surfaces as an explicit arithmetic input, we prove that an effective zero-cycle of degree five on a cubic threefold implies the existence of a rational point. The construction associates to a spanning quintic point an auxiliary cubic in a space of tangential rational quintics. A determinantal curve of degree ten contains five conjugate lines; the residual effective one-cycle has degree five and is supported on the auxiliary cubic. The surface input produces a point on that cubic, and a residue identity transfers it to the original threefold. The nonemptiness calculation is accompanied by a finite exact certificate. Degree eight follows from classical rational-normal-curve interpolation, Balestrieri's degree-four reduction, and Ma's degree-seven reduction. The argument does not establish the conjecture for arbitrary zero-cycles of degree one.

1 Introduction

Let $X \subset \mathbf{P}_k^4$ be a smooth cubic threefold over a field k of characteristic zero. The conjecture of Cassels and Swinnerton-Dyer asks whether

$$X \text{ has a zero-cycle of degree } 1 \implies X(k) \neq \emptyset. \quad (1.1)$$

Equivalently, the existence of a point over a finite extension of degree prime to three should imply the existence of a k -point. Indeed, a line meeting X properly supplies an effective zero-cycle of degree three. The equivalence then follows by taking degrees and applying Bézout's identity. Coray's work [Cor76] is a fundamental reference for this conjecture and for geometric degree-reduction methods.

Here and throughout, a zero-cycle is an integral linear combination of closed points; it is *effective* when its coefficients are nonnegative. This distinction is essential. Our small-degree hypotheses concern effective zero-cycles, not arbitrary representatives of the same degree in $\mathrm{CH}_0(X)$.

1.1 Previous results and their dimensions

Quadratic descent is elementary for cubic hypersurfaces of every dimension: the line through two conjugate points either lies in the hypersurface or has a residual rational intersection point. Coray proved the full conjecture over local fields [Cor76, Theorem 4.7], so the issue here is descent over general characteristic-zero fields. For threefolds, Ma proved the following reduction [Ma21, Theorem 3.6]:

$$\text{a closed point of degree } 7 \implies \text{a closed point of degree } 1, 2, 4, \text{ or } 5. \quad (1.2)$$

His argument develops Coray's method of residual intersections with rational normal curves and deals with configurations not in general position. Ma also proved [Ma21, Proposition 6.2] that

$\mathrm{Sym}^7 X$ is birational to $\mathrm{Sym}^5 X \times \mathbf{P}_k^6$. The degree-eight statement in [Ma21, Theorem 3.6] concerns *cubic fourfolds in $\mathbf{P}^5$* ; it reduces a degree-eight point to degrees 1, 2, 4, 5, 7. It is not the degree-eight rational-point assertion for threefolds proved below.

Balestrieri subsequently proved a dimension-independent statement [Bal24, Theorem 3.8]: if a cubic hypersurface has a point over a *simple* extension of degree four, then it has a point over an extension of degree one or five. Thus, in characteristic zero,

$$\text{a closed point of degree 4} \implies \text{a closed point of degree 1 or 5.} \quad (1.3)$$

Together, (1.2) and (1.3) show that degree-five descent is the remaining arithmetic ingredient for the degree-four and degree-seven conclusions considered here. Neither of these published reductions alone concludes that a rational point exists.

For comparison, the surface theory is substantially more developed. Coray reduced prime-to-three degrees on smooth cubic surfaces over perfect fields to 1, 4, 10 [Cor76, Theorem 7.1]; Ma extended the reduction to regular cubic surfaces over arbitrary fields [Ma21, Theorem 2.6]. Colliot-Thélène revisited Coray’s method using generization, specialization, and Bertini theorems, and developed related effectivity statements [CT20]. Voisin’s rank-two-vector-bundle method reduces the surface problem in characteristic zero to degrees 1 and 4 [Voi26, Theorem 1.5]. These are surface results, not dimension-three degree bounds. In another direction, Voisin proves unirationality of the third symmetric product of a cubic hypersurface of dimension at least two [Voi26, Theorem 1.1]; this concerns degree-three zero-cycles rather than prime-to-three descent.

1.2 Arithmetic input and main results

The arithmetic input in this paper is the following statement from the companion cubic-surface work [Surf]. We state it separately to make the dependence of the threefold argument precise.

Surface input 1.1 (S_5). For every characteristic-zero field K , a smooth cubic surface $S \subset \mathbf{P}_K^3$ with an effective zero-cycle of degree five has a K -point.

The statement is used as an input, not reproved here. In particular, it must be available over the function fields that occur in the argument. It is stronger than the published surface reductions just recalled and is not attributed to those reductions.

Theorem 1.2. Assume (S_5). Let k be a field of characteristic zero and let $X \subset \mathbf{P}_k^4$ be a smooth cubic threefold. If X has an effective zero-cycle of degree 5 or 8, then $X(k) \neq \emptyset$.

Corollary 1.3. Under the same surface input, an effective zero-cycle of any degree in $\{1, 2, 4, 5, 7, 8\}$ on a smooth cubic threefold implies a rational point.

The principal construction is the degree-five case. The degree-eight case is a consequence of this case and the interpolation method already used by Coray and Ma; it should not be regarded as a second independent descent mechanism. The degree-four case also follows directly from the corresponding surface case by taking a generic hyperplane section and using the Lang–Nishimura theorem; we give this argument in Proposition 6.1.

The new geometric step is the construction of an auxiliary cubic Y carrying an effective one-cycle of degree five, together with a transfer from $Y(k)$ to $X(k)$. The degree of the determinantal curve is supplied by the Eagon–Northcott resolution [EN62], not by a new determinantal formula. The uses of specialization and symmetric powers are standard; we use the theorem of Lang and Nishimura [Lan54, Nis55], in the form recorded in [Poo17, Theorem 3.6.11]. Only (S_5) is required from the surface work: a deformation argument below handles singular hyperplane sections, so neither a Pfaffian presentation of X nor a separate arithmetic theorem for singular cubic surfaces is assumed.

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2 Preliminary reductions

All fields in the sequel have characteristic zero. Write $X^{(d)}$ for $\text{Sym}^d X$. The degree of a one-cycle in projective space means its intersection number with a hyperplane. Proper pushforward of zero-cycles preserves their degree over the ground field.

Lemma 2.1 (Quadratic descent). *Let $V \subset \mathbf{P}_k^n$ be the zero locus of a cubic form. If V has an effective zero-cycle of degree two, then $V(k) \neq \emptyset$.*

Proof. Only a closed point of degree two needs consideration. Its two conjugate geometric points span a k -line. If the line is contained in V , it has k -points. Otherwise its degree-three intersection divisor with V contains the given degree-two point, leaving an effective degree-one divisor. $\square$

Lemma 2.2 (Specialization of a point). *Let F_z be a homogeneous form with coefficients in $k[z]$ and suppose that $F_0 \neq 0$. If $V(F_z)$ has a point over $k(z)$, then $V(F_0)$ has a point over k .*

Proof. Scale the homogeneous coordinates of the given point so that they belong to the discrete valuation ring $k[z]_{(z)}$ and at least one is a unit. Reducing the equation modulo z gives a nonzero vector satisfying $F_0 = 0$. This is the valuative criterion for properness in coordinates. $\square$

Lemma 2.3 (Cubics through a projective frame). *Let $Z \subset \mathbf{P}_k^3$ be a reduced degree-five subscheme whose geometric points have no four coplanar. The space of cubic forms vanishing on Z contains a nonempty k -open subset defining smooth cubic surfaces.*

Proof. It suffices first to check geometric nonemptiness. Over $\bar{k}$ the five points can be written as the four coordinate vertices and $[1 : 1 : 1 : 1]$. The quadrics through them are

$$\sum_{0 \leq i < j \leq 3} c_{ij} x_i x_j, \quad \sum_{i < j} c_{ij} = 0.$$

Their common zeros satisfy that all six products $x_i x_j$ are equal. If their common value is zero the point is a coordinate vertex; otherwise all four coordinates are equal. Thus the base locus is precisely Z . At a coordinate vertex the differentials of these quadrics span the cotangent space; at $[1 : 1 : 1 : 1]$ the same assertion follows from the differences $x_0 x_1 - x_0 x_2$, $x_0 x_1 - x_0 x_3$, and $x_0 x_2 - x_1 x_2$.

Multiplying the quadrics by linear forms shows that the cubic system is base-point-free away from Z , and that its differentials span the cotangent space at each point of Z . Bertini's theorem in characteristic zero gives a general cubic smooth away from Z ; the finitely many additional nonsingularity conditions at Z are nonempty open conditions. Smoothness is open in the parameter space of cubic forms and is defined over k . Finally, a nonempty open subset of an affine space over the infinite field k has a k -point. $\square$

Lemma 2.4 (A quintic cycle in a proper linear subspace). *Assume (S_5) . Let $V \subset \mathbf{P}_k^4$ be the zero locus of a cubic form. If V has an effective degree-five zero-cycle whose support is contained in a k -hyperplane, then $V(k) \neq \emptyset$. No smoothness assumption on V or on the hyperplane section is needed.*

Proof. If the cycle contains a point of degree one or two, the assertion follows from Lemma 2.1. Otherwise it is a single closed point $Z = \text{Spec } E$ of degree five.

Suppose first that its linear span is a k -space $\mathbf{P}^3$. Choose a k -linear form nonvanishing at Z and normalize the five geometric representatives accordingly. They have a unique linear relation up to scalar. The subset of its zero coefficients is invariant under Galois, which acts transitively on the five representatives. Hence no coefficient is zero, and any four representatives are independent.

Let f_0 be the cubic restricted to this $\mathbf{P}^3$. If $f_0 = 0$, there is nothing to prove. Otherwise Lemma 2.3 supplies a smooth cubic f_1 through Z . The generic member $f_z = (1 - z)f_0 + zf_1$ is smooth and contains $\text{Spec } E(z)$, a closed point of degree five over $k(z)$. Apply (S_5) over $k(z)$ and then Lemma 2.2.

If the span of Z has dimension at most two, choose a k -plane containing it. There is a nonzero conic through Z , since conic forms have dimension six and Z imposes at most five linear conditions. If its intersection with the restricted cubic is proper, the degree-six intersection cycle contains $[Z]$ and leaves an effective zero-cycle of degree one. If the two equations have a common factor over k , a linear common factor is already a k -line on V ; a quadratic common factor leaves a k -linear factor of the cubic, again giving such a line. A zero cubic restriction is immediate. $\square$

Lemma 2.5 (An effective one-cycle on a cubic). *Assume (S_5) . Let $Y \subset \mathbf{P}_k^4$ be the zero locus of a cubic form. If Y contains an effective one-cycle of degree five, then $Y(k) \neq \emptyset$.*

Proof. Choose a k -hyperplane meeting every component of the cycle properly. Its intersection with the cycle is effective of degree five and is supported on Y in that hyperplane. Apply Lemma 2.4. The assertion is immediate if the cubic form is identically zero. $\square$

Lemma 2.6 (Symmetric powers and specialization). *Let X be a smooth geometrically integral proper k -variety. If $X^{(d)} \dashrightarrow X$ is a rational map, then a separable closed point of degree d on X implies a k -point on X .*

Proof. The closed point defines a k -point of the open configuration locus in $X^{(d)}$ parametrizing d distinct geometric points. This open is smooth: it is the quotient of the distinct-point locus in X^d by the free action of the symmetric group. The Lang–Nishimura theorem, applied at this smooth point and with proper target X , gives the assertion. No smoothness of the entire symmetric power is asserted or required. $\square$

3 Tangential rational quintics and an auxiliary cubic

We now treat a closed point $Z = \text{Spec } E \subset X$ of degree five which spans $\mathbf{P}^4$. The nonsmooth cubic loci occurring as auxiliary objects will be allowed throughout. After normalizing the five geometric representatives of Z by a k -linear form, they form a semilinearly permuted basis. Their ambient vector space is thereby identified with the descended permutation space E . We use split coordinates indexed by $i = 0, \dots, 4$. All constructions below commute with the semilinear permutation of these indices and therefore descend to k .

Since the marked points are coordinate vertices, the cubic has the form

$$F(x) = \sum_{i \neq j} a_{ij} x_i^2 x_j + \sum_{i < j < r} b_{ijr} x_i x_j x_r, \quad A = (a_{ij}), \quad a_{ii} = 0. \quad (3.1)$$

The subscript on b_{ijr} will always denote an unordered triple. Choose a primitive element $\alpha \in E$ with conjugates α_i and put

$$D = \text{diag}(\alpha_i), \quad B_{ii} = 0, \quad B_{ij} = \frac{a_{ij}}{\alpha_i - \alpha_j}, \quad L = B^{-1}DB. \quad (3.2)$$

Until Section 5, assume that A and B are invertible. The identity

$$A = DB - BD \quad (3.3)$$

implies, for a variable $h \in E$, that

$$\mu = (D - L)h = -B^{-1}Ah, \quad Ah + B\mu = 0. \quad (3.4)$$

Consider the rational vector

$$r_h(T) = h + (T - D)^{-1}\mu = (T - D)^{-1}(T - L)h. \quad (3.5)$$

The conjugacy of D and L gives

$$\det((T - D)^{-1}(T - L)) = 1. \quad (3.6)$$

Let

$$p(s, t) = \prod_i (s - \alpha_i t), \quad p_i(s, t) = \prod_{j \neq i} (s - \alpha_j t).$$

Clearing denominators in (3.5) gives five binary quintics

$$\gamma_{h,i}(s, t) = h_i p(s, t) + \mu_i t p_i(s, t). \quad (3.7)$$

The second identity in (3.4) is the linear tangency condition at the five marked points. The following calculation also controls the residual intersection.

For $j \neq i$ write

$$u_{ij} = h_j + \frac{\mu_j}{\alpha_i - \alpha_j}.$$

Define quadratic forms

$$q_i(h) = \sum_{j \neq i} a_{ji} u_{ij}^2 + \sum_{\substack{j < r \\ j, r \neq i}} b_{ijr} u_{ij} u_{ir} - \mu_i \sum_{j \neq i} \frac{a_{ij} \mu_j}{(\alpha_i - \alpha_j)^2} \quad (3.8)$$

and the cubic form

$$G(h) = F(h) - \sum_i h_i q_i(h), \quad Y = V(G) \subset \mathbf{P}(E). \quad (3.9)$$

Both G and the pairing $\sum_i q_i(h) x_i$ are defined over k .

Proposition 3.1 (Residue identity). *One has*

$$F(r_h(T)) = \sum_i q_i(h) r_{h,i}(T) + G(h). \quad (3.10)$$

Equivalently, the homogeneous forms satisfy

$$F(\gamma_h) = p^2 \left(\sum_i q_i(h) \gamma_{h,i} + G(h) p \right). \quad (3.11)$$

Proof. Write

$$F = x_i^2 f_{1,i}(x_{\neq i}) + x_i f_{2,i}(x_{\neq i}) + f_{3,i}(x_{\neq i}),$$

where

$$f_{1,i}(y) = \sum_{j \neq i} a_{ij} y_j, \quad f_{2,i}(y) = \sum_{j \neq i} a_{ji} y_j^2 + \sum_{\substack{j < r \\ j, r \neq i}} b_{ijr} y_j y_r.$$

There is no triple pole of $F(r_h(T))$ at $T = \alpha_i$, because F has no pure cubes. Its possible double-pole coefficient is

$$\mu_i^2 f_{1,i}(u_i) = \mu_i^2 (Ah + B\mu)_i = 0.$$

Its simple residue is

$$\mu_i f_{2,i}(u_i) - \mu_i^2 \sum_{j \neq i} \frac{a_{ij} \mu_j}{(\alpha_i - \alpha_j)^2} = \mu_i q_i(h).$$

The possible additional term $2\mu_i h_i f_{1,i}(u_i)$ vanishes by the same tangency equation. Subtracting $\sum_i q_i(h) r_{h,i}(T)$ therefore removes every finite pole. The difference is bounded at infinity, where its value is $G(h)$. This proves (3.10); homogenization after clearing denominators gives (3.11). $\square$

Proposition 3.2 (Transfer). *Assume (S₅) and $\det A \det B \neq 0$. Then $Y(k) \neq \emptyset$ implies $X(k) \neq \emptyset$.*

Proof. Represent a point of $Y(k)$ by $0 \neq h \in E$. The invertibility of $-B^{-1}A$ implies $\mu \neq 0$. Since E is a *field*, all its conjugates μ_i are nonzero.

The forms (3.7) have no common zero. Away from the roots of p this follows from (3.6). At $[\alpha_i : 1]$ their only nonzero coordinate is $\gamma_{h,i}(\alpha_i, 1) = \mu_i p_i(\alpha_i, 1)$, and at infinity the vector of coordinates is $h \neq 0$. Thus they define a k -morphism

$$\gamma_h : \mathbf{P}_k^1 \longrightarrow \mathbf{P}_k^4, \quad \gamma_h^* \mathcal{O}_{\mathbf{P}^4}(1) \simeq \mathcal{O}_{\mathbf{P}^1}(5).$$

Its image spans $\mathbf{P}^4$, since it contains the five marked vertices.

If every $q_i(h)$ vanishes, (3.11) puts this image in X , and evaluation at a k -point of $\mathbf{P}^1$ suffices. Otherwise

$$Q_h(s, t) = \sum_i q_i(h) \gamma_{h,i}(s, t)$$

is a nonzero binary quintic: a nonzero linear form cannot vanish on a spanning image. As $G(h) = 0$, the divisor $V(Q_h)$ maps to X by (3.11). Its pushforward is an effective degree-five zero-cycle supported in the k -hyperplane $\sum_i q_i(h) x_i = 0$. Apply Lemma 2.4. $\square$

Remark 3.3. The transfer works for every point of $Y(k)$, not merely for points in a chosen open subset. The use of the field E is essential here: in a split *algebra*, a nonzero vector can have zero coordinates. Split parameters will be used below only to certify geometric nonemptiness, not to justify the arithmetic transfer.

4 A determinantal curve and its residual cycle

Consider the 5×3 matrix of linear forms

$$N(h) = [h \quad (D + L)h \quad DLh], \quad \mathcal{C} = V(I_3(N)) \subset \mathbf{P}(E). \quad (4.1)$$

Here $I_3(N)$ denotes the ideal of its maximal minors. The matrix, and hence the determinantal scheme, descend to k .

Lemma 4.1 (Exceptional lines). *For each i the vector space*

$$W_i = \ker((D - \alpha_i I)(L - \alpha_i I)) \quad (4.2)$$

has dimension two. Its projectivization $\ell_i = \mathbf{P}(W_i)$ is a line on $\mathcal{C}$, and the sum of the five conjugate lines descends to k .

Proof. The map $L - \alpha_i I$ has a one-dimensional kernel. Its image is the hyperplane annihilated by $e_i^\top B$, which contains e_i since $B_{ii} = 0$. Its inverse image of the line $\bar{k}e_i$ is therefore two-dimensional; this is precisely W_i . For $h \in W_i$ one has

$$(\alpha_i^2 I - \alpha_i(D + L) + DL)h = 0,$$

a nonzero relation among the columns of $N(h)$. Galois compatibility gives the last assertion. $\square$

Lemma 4.2 (The nonexceptional locus). *Every geometric point of $\mathcal{C}$ outside $\bigcup_i \ell_i$ lies on Y .*

Proof. Work over $\bar{k}$. A point $[h] \in \mathcal{C}$ admits a nonzero relation

$$(aI + b(D + L) + cDL)h = 0. \quad (4.3)$$

Suppose $b^2 - ac = 0$. Then $c \neq 0$, since otherwise (4.3) would force $h = 0$. Rescaling the relation gives $(a, b, c) = (\tau^2, -\tau, 1)$, and its matrix is $(D - \tau I)(L - \tau I)$. Its determinant is $\prod_i (\alpha_i - \tau)^2$. Thus $\tau = \alpha_i$ for some i , and $[h] \in \ell_i$.

Consequently, outside the five lines a relation as in (4.3) has $b^2 - ac \neq 0$. Let $t_+, t_- \in \mathbf{P}^1(\bar{k})$ be the distinct roots of $cT^2 + 2bT + a$; if $c = 0$, one root is infinity. Set $l = Lh$. The component equations of (4.3) are

$$(a + b\alpha_i)h_i + (b + c\alpha_i)l_i = 0. \quad (4.4)$$

When neither root is α_i , substituting into $r_i(T) = (Th_i - l_i)/(T - \alpha_i)$ gives $r_i(t_+) + r_i(t_-) = 0$. This also holds with a root at infinity, using $r_i(\infty) = h_i$.

We must include the possibility that a root equals α_i . Equation (4.4), together with its being a simple root, then gives $l_i = \alpha_i h_i$, or $\mu_i = 0$. The apparent pole cancels and $r_i(T) = h_i$ is constant. Let I_0 be the set of such indices; it has at most two elements. For $u = \sum_{i \in I_0} h_i e_i$ and some v supported outside I_0 , the regularized values satisfy

$$r_h(t_+) = u + v, \quad r_h(t_-) = u - v. \quad (4.5)$$

We claim that

$$\frac{F(u + v) + F(u - v)}{2} = \sum_{i \in I_0} h_i q_i(h). \quad (4.6)$$

If I_0 is empty, both sides vanish by oddness. If $I_0 = \{i\}$, then $q_i = f_{2,i}(v)$, since $\mu_i = 0$; the even part of $F(h_i e_i + v)$ is exactly $h_i f_{2,i}(v)$.

Finally let $I_0 = \{i, j\}$, with $t_+ = \alpha_i$ and $t_- = \alpha_j$. Then

$$q_i = f_{2,i}(h_j e_j + v), \quad q_j = f_{2,j}(h_i e_i - v).$$

In $h_i q_i + h_j q_j$ the terms $b_{ijr} h_i h_j v_r$ cancel. The constant terms give $F(h_i e_i + h_j e_j)$, and the remaining terms are precisely those linear in u and quadratic in v . This is the even part on the left of (4.6).

Evaluate (3.10) at the two regularized values (4.5) and add. By (4.6) the left side is $2 \sum_{i \in I_0} h_i q_i$, while the right side is that same expression plus $2G(h)$. Therefore $G(h) = 0$. $\square$

Proposition 4.3 (The residual degree-five cycle). *Suppose that $\mathcal{C}$ has codimension three in $\mathbf{P}(E)$, and that the five lines ℓ_i are distinct components of multiplicity one. Then*

$$\Xi = [\mathcal{C}] - \sum_{i=0}^4 [\ell_i] \quad (4.7)$$

is an effective k -one-cycle of degree five supported on Y .

Proof. For a 5×3 matrix with maximal-minor ideal of the expected codimension, the Eagon–Northcott resolution [EN62] is

$$0 \longrightarrow R(-5)^6 \longrightarrow R(-4)^{15} \longrightarrow R(-3)^{10} \longrightarrow R \longrightarrow R/I_3(N) \longrightarrow 0, \quad (4.8)$$

where $R = \bar{k}[h_0, \dots, h_4]$. Thus $\mathcal{C}$ is Cohen–Macaulay and pure of dimension one. Its Hilbert polynomial is $10m - 5$, so its degree is ten. Subtracting the five distinct multiplicity-one line components leaves an effective cycle of degree five. Lemma 4.2 puts the generic point of each remaining component on Y , hence its whole support on Y . Both terms of (4.7) are Galois invariant, so the cycle descends to k . $\square$

Remark 4.4. The cycle Ξ need not be smooth, integral, or reduced. Its effectivity, degree, and support are the only properties used. Likewise, no smoothness assertion about the auxiliary cubic Y is needed.

5 Nonempty parameters and quintic descent

Lemma 5.1. *Fix a spanning quintic closed point $Z = \operatorname{Spec} E \subset \mathbf{P}_k^4$. In the parameter space of cubic forms through Z and primitive elements of E , the following conditions hold on a geometrically nonempty k -open:*

$$\det A \det B \neq 0, \quad \dim \mathcal{C} = 1, \quad \ell_0, \dots, \ell_4 \text{ distinct and of multiplicity one in } \mathcal{C}. \quad (5.1)$$

Proof. The parameter space is $V_Z \times \operatorname{Prim}(E)$, where $V_Z = H^0(\mathbf{P}^4, \mathcal{I}_Z(3))$ has dimension $35 - 5 = 30$, and $\operatorname{Prim}(E)$ is a nonempty open in the five-dimensional vector space E . After a splitting extension, its coordinates are the twenty a_{ij} , ten b_{ijr} , and five pairwise distinct α_i .

Invertibility is open. The dimension condition is open by upper semicontinuity for the projective family of maximal-minor loci; the exceptional lines ensure that the dimension cannot be less than one. On the open where B is invertible and the α_i are distinct, the spaces W_i form rank-two vector bundles on the split parameter space. Require each corresponding line to contain a point where N has rank two and the Jacobian of its ten maximal minors has rank three. This is open on the parameter base: the locus of such points is open in the associated projective-line bundle, whose projection is an open map. It ensures multiplicity one. Distinctness of the lines is open as well. These conditions are invariant under permutation of the conjugates and descend.

Appendix A gives an explicit simultaneous witness over $\mathbf{F}_{101}$ and verifies it by exact linear algebra. The formulas and these open conditions are defined on an open subset of affine space over $\mathbf{Z}$, obtained by inverting the indicated denominators. This parameter space is integral, and the denominators are nonzero at the witness. A nonempty open containing the witness therefore meets the characteristic-zero generic fiber. It follows that the good locus is geometrically nonempty in characteristic zero. After splitting E , the parameter space for the prescribed Z is the same split space, so the conclusion holds for every such E/k . $\square$

Proposition 5.2. *Assume (S_5) . A cubic with a spanning quintic point and parameters in the open of Lemma 5.1 has a rational point.*

Proof. Proposition 4.3 supplies the effective degree-five one-cycle Ξ on Y . Lemma 2.5 gives $Y(k) \neq \emptyset$, and Proposition 3.2 transfers a point to the original cubic. $\square$

Theorem 5.3 (Degree five). *Assume (S_5) . If a smooth cubic threefold X/k has an effective zero-cycle of degree five, then $X(k) \neq \emptyset$.*

Proof. An effective cycle of degree five either contains a point of degree one or two, or consists of a single closed point $Z = \text{Spec } E$ of degree five. The first alternatives follow from Lemma 2.1. If Z does not span $\mathbf{P}^4$, use Lemma 2.4.

Suppose that Z spans $\mathbf{P}^4$, and let F_0 define X . Lemma 5.1 gives a geometrically nonempty k -open in the affine parameter space $V_Z \times \text{Prim}(E)$. Since k is infinite, choose a good k -point (F_1, α_1) and any primitive element $\alpha_0 \in E$. Put

$$F_z = (1 - z)F_0 + zF_1, \quad \alpha_z = (1 - z)\alpha_0 + z\alpha_1. \quad (5.2)$$

The line (5.2) meets the good open at $z = 1$ and hence has good generic point. The algebra $E \otimes_k k(z) = E(z)$ is still a field of degree five. Proposition 5.2, applied over $k(z)$, gives a $k(z)$ -point of $V(F_z)$. Specialization by Lemma 2.2 gives the required point of $V(F_0)$. $\square$

Remark 5.4. The last step specializes a point of the original cubic family, not a component or point of a limiting auxiliary curve. Nothing requires the auxiliary construction to extend over $z = 0$.

6 Degrees four, seven, and eight

6.1 Degree four and hyperplane sections

Proposition 6.1 (Passage from surfaces in degree four). *If effective degree-four descent holds for smooth cubic surfaces over all characteristic-zero extensions of k , then it holds for smooth cubic threefolds over k . In particular it holds under (S_5) .*

Proof. Let X be a smooth cubic threefold and set $K = k(X^{(4)})$. The generic quartet spans a hyperplane whose section $S \subset X_K$ is a smooth cubic surface. To check this general-position assertion, work over $\bar{k}$, choose a smooth hyperplane section, and choose four points on it spanning that hyperplane. These properties are open.

The surface S has the universal effective degree-four cycle, so the surface degree-four statement gives $S(K) \neq \emptyset$. This gives a rational map $X^{(4)} \dashrightarrow X$. Lemma 2.6 handles an arbitrary closed point of degree four. A reducible effective four-cycle contains a point of degree one or two.

Under (S_5) , the needed surface degree-four statement follows from Balestrieri's theorem (1.3), applied to a cubic surface: a point over a degree-five extension is either rational already or is a closed point of degree five. Alternatively, Balestrieri's theorem may be applied directly to X , followed by Theorem 5.3. $\square$

Remark 6.2. The companion surface statement that a degree-one zero-cycle implies a rational point also gives the surface degree-four case immediately: an effective four-cycle minus a degree-three line section has degree one. Thus the first paragraph of Proposition 6.1 is precisely the direct reduction of the threefold degree-four case to the surface case.

6.2 Ma's degree-seven reduction

Corollary 6.3 (Degree seven). *Assume (S₅). An effective zero-cycle of degree seven on a smooth cubic threefold implies a rational point.*

Proof. For a closed point of degree seven, apply Ma's Theorem 3.6 [Ma21]. The resulting degrees 1, 2, 4, 5 are covered by Lemma 2.1, Proposition 6.1, and Theorem 5.3. For a reducible effective seven-cycle, some support degree is prime to three and smaller than seven, hence is 1, 2, 4, or 5. $\square$

6.3 Lifting eight points and interpolating a quintic

Lemma 6.4 (Lifting a configuration). *Let K be an infinite field of characteristic zero, and let $Z \subset \mathbf{P}_K^4$ be a reduced degree-eight subscheme whose geometric points are in linear general position. There is a lift $\tilde{Z} \subset \mathbf{P}_K^5$, in linear general position, under projection from a K -point $o \in \mathbf{P}^5$. Through $\tilde{Z}$ there passes a K -defined rational normal quintic C , whose underlying curve is isomorphic to $\mathbf{P}_K^1$.*

Proof. Choose coordinates so that x_0 does not vanish at any point of Z , and normalize $x_0 = 1$. Put $E = H^0(Z, \mathcal{O}_Z)$, an étale algebra of dimension eight over K . An additional coordinate $\beta \in E$ gives a lift to $\mathbf{P}_K^5$, with center $o = [0 : 0 : 0 : 0 : 0 : 1]$.

After splitting E , write the eight lifted points as (v_i, β_i) , with $v_i \in \bar{K}^5$. Every five v_i are independent. For any subset of six points, the determinant of the six lifted vectors is a nonzero linear form in their β_i : its coefficients are the five-by-five minors of the corresponding original vectors, all nonzero. Avoiding these finitely many hyperplanes makes every six lifted vectors independent. The resulting nonempty open is defined over K in the affine space E , so it has a K -point.

The classical interpolation theorem gives a unique rational normal quintic through eight points in linear general position in $\mathbf{P}^5$; see the descended form in [Ma21, Proposition 3.3]. Uniqueness implies that the curve descends to K . Its underlying curve is smooth, projective, and geometrically of genus zero. The line bundle

$$\mathcal{O}_C(1) \otimes \omega_C^{\otimes 2}$$

has degree $5 - 4 = 1$. Riemann–Roch gives a nonzero section and hence an effective degree-one divisor. Thus $C(K) \neq \emptyset$ and $C \simeq \mathbf{P}_K^1$. $\square$

Theorem 6.5 (Degree eight). *Assume (S₅). If a smooth cubic threefold X/k has an effective zero-cycle of degree eight, then $X(k) \neq \emptyset$.*

Proof. First work over $K = k(X^{(8)})$ with the universal reduced degree-eight subscheme $Z \subset X_K$. Its geometric points are in linear general position in $\mathbf{P}^4$: this is a nonempty open condition on tuples of points of the nondegenerate variety X .

Apply Lemma 6.4. Project the resulting rational normal quintic $C \subset \mathbf{P}_K^5$ back to $\mathbf{P}_K^4$. If the projection center does not lie on C , the five coordinate sections are base-point-free of degree five. If the center lies on C , their common zero has multiplicity exactly one, since C is smoothly embedded; removing it gives a base-point-free system of degree four. In both cases the projection extends to a morphism

$$\gamma : C \simeq \mathbf{P}_K^1 \longrightarrow \mathbf{P}_K^4, \quad m = \deg \gamma^* \mathcal{O}_{\mathbf{P}^4}(1) \in \{4, 5\}.$$

The marked subscheme $\tilde{Z}$ avoids the center and maps isomorphically to Z .

If $\gamma(C) \subset X_K$, evaluation at a K -point of C gives a point of X_K . Otherwise the cubic pullback has an effective zero divisor of degree $3m$ containing $\tilde{Z}$. Its residual divisor is effective of degree

$$3m - 8 \in \{4, 7\}. \tag{6.1}$$

Pushing it to X_K gives an effective cycle of that degree. Proposition 6.1 or Corollary 6.3 therefore gives $X(K) \neq \emptyset$.

We have constructed a rational map $X^{(8)} \dashrightarrow X$. Lemma 2.6 treats every closed point of degree eight, without any position assumption. If the original effective eight-cycle is not a single such point, it contains a point of degree prime to three in $\{1, 2, 4, 5, 7\}$, already covered. $\square$

Proof of Theorem 1.2 and Corollary 1.3. Theorem 1.2 is Theorems 5.3 and 6.5. The additional degrees follow from Lemma 2.1, Proposition 6.1, and Corollary 6.3. $\square$

7 Scope of the conclusions

Under (S_5) , the results also give rational maps

$$X^{(d)} \dashrightarrow X, \quad d \in \{4, 5, 7, 8\}.$$

For $d = 5, 7$ this follows by applying the corresponding result to the universal cycle over $k(X^{(d)})$; the constructions for $d = 4, 8$ were used above. The maps are existence statements and are not asserted to be canonical.

There is a useful reason not to isolate degree eight as a separate arithmetic input. If z_5 is an effective degree-five cycle and c_3 is a degree-three line-section cycle, then $z_5 + c_3$ is an effective eight-cycle. Thus the uniform effective degree-eight assertion implies the degree-five assertion. Conversely, the arguments of Section 6 show that uniform degree-five descent for smooth cubic threefolds, combined with Balestrieri's and Ma's reductions, implies effective degree-eight descent. The degree-eight result is therefore a corollary of the quintic case in exactly this sense.

Finally, the presence of an arbitrary zero-cycle of degree one gives some closed point of degree prime to three, but not necessarily of any of the degrees treated here. No reduction from all prime-to-three degrees to $\{1, 2, 4, 5, 7, 8\}$ is proved. Likewise, a cycle with negative coefficients of degree five or eight cannot be substituted for an effective cycle in the hypotheses: together with a degree-three cycle, either such formal degree condition is equivalent to index one. The passage from index one to a rational point on an arbitrary smooth cubic threefold remains outside the conclusions of this paper.

A An exact nonemptiness certificate

We supply a witness for Lemma 5.1. The calculation is only used to prove that the genericity conditions define a nonempty open; it makes no assertion about rational points over arbitrary fields. All indices in this appendix are zero-based.

Over $\mathbf{F}_{101}$ take

$$(\alpha_i) = (0, 1, 2, 3, 4), \quad A = \begin{pmatrix} 0 & 2 & 3 & -1 & 1 \\ 1 & 0 & 2 & 1 & -1 \\ 3 & 3 & 0 & 0 & 1 \\ 1 & 3 & -2 & 0 & 1 \\ 0 & -2 & 1 & 0 & 0 \end{pmatrix}. \quad (\text{A.1})$$

Define D, B, L, N by (3.2) and (4.1). Then

$$\det A = 20, \quad \det B = 29 \quad \text{in } \mathbf{F}_{101}.$$

The coefficients b_{ijr} are irrelevant to these genericity conditions; they may all be zero. In particular, no smoothness test for Y is necessary.

A.1 Expected dimension

Restrict the ten cubic maximal minors of N to the plane with matrix

$$U = \begin{pmatrix} 13 & 11 & 93 \\ 32 & 85 & 76 \\ 24 & 73 & 17 \\ 48 & 35 & 76 \\ 5 & 10 & 81 \end{pmatrix}, \quad h = U(y_0, y_1, y_2)^\top. \quad (\text{A.2})$$

Order the row triples of N lexicographically. For each restricted minor, multiply by each of the six degree-two monomials, ordered in descending lexicographic order. Express the resulting sixty quintics in the twenty-one degree-five monomials, also in descending lexicographic order. Let M be this 60×21 coefficient matrix. The square submatrix with row indices

$$0, 1, 2, \dots, 19, 36 \quad (\text{A.3})$$

and all columns has determinant 74 in $\mathbf{F}_{101}$. Thus M has rank 21. The restricted minor ideal contains every quintic in three variables and consequently has no projective zero over $\overline{\mathbf{F}}_{101}$. Hence the determinantal locus misses this plane. A projective subscheme of dimension at least two in $\mathbf{P}^4$ meets every plane, so $\dim \mathcal{C} \leq 1$. The five lines give equality.

A.2 Multiplicity and distinctness of the lines

At the following points of the respective ℓ_i , the ranks are as shown. The Jacobian is that of all ten maximal minors in the five coordinates h_j .

i	h	rank $N(h)$	Jacobian rank
0	(76, 72, 96, 1, 1)	2	3
1	(31, 43, 95, 1, 1)	2	3
2	(93, 1, 29, 85, 1)	2	3
3	(1, 67, 22, 1, 1)	2	3
4	(77, 24, 94, 1, 1)	2	3

Thus $\mathcal{C}$ is a smooth curve at each displayed projective point, so the corresponding line component has multiplicity one. For every pair $i \neq j$, concatenating bases of W_i and W_j gives a matrix of rank four. In particular the five lines are distinct (indeed disjoint) for this witness.

A.3 Reproduction

The accompanying script `verify_determinantal_certificate.py` uses only the Python standard library and exact arithmetic modulo 101. It reconstructs B, L , the minor coefficient matrix, the indicated nonzero square minor, and all line and Jacobian tests. Running

```
python verify_determinantal_certificate.py
```

writes the data to `determinantal_certificate.json`. A successful local run is recorded in `verification_output.txt`. The general residue identity, the nonexceptional-locus lemma, Galois descent, and specialization are proved symbolically in the text. The certificate verifies none of the arithmetic content of (S_5) .

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